

50-Point Architectural Safety Audit

The Technical Due Diligence Standard for the Riviera Maya

Objective Reality vs. Marketing Hype

In the Riviera Maya, luxury is often a coat of paint over structural shortcuts. This audit is designed to identify the second-order effects of poor construction before they become your financial liability. We prioritize evidence over sales brochures.

I. Structural Integrity & Foundation (The Karst Reality)

- **Mecánica de Suelos:** Verification of the soil mechanics study.
- **Rebar Density & Grade:** Checking for oxidation in salt air environments.
- **Foundation Depth:** Verification that foundation reaches solid bedrock.
- **Structural Expansion Joints:** Capability to handle thermal expansion.
- **Concrete Compression Tests:** Review of 'ensayos de concreto' (lab results).
- **Cantilever Stress Marks:** Hairline fracture inspection on balconies.
- **Roof Load Capacity:** Structural integrity for rooftop pools.
- **Masonry Bond:** Integration of columns (castillos) and beams (dalas).

II. Water Management & Moisture Protection

- **Impermeabilización (Roofing):** TPO/asphalt membrane vs. cheap paint.
- **Negative-Side Waterproofing:** Groundwater seepage in basement/parking.
- **Window Flashings:** Seal quality between frames and masonry.
- **Balcony Drainage:** Slope calculation (pooling away from glass).
- **AC Condensation Lines:** Gravity vs. pump-fed drainage routing.
- **Humidity Barrier:** Moisture protection behind bathroom tiling.
- **Septic/Sump Pump Redundancy:** Emergency pumps for elevator pits.
- **Efflorescence Check:** White salt powder signs of rising dampness.

III. Electrical & Mechanical Systems

- **Grounding System (Tierras):** Physical copper grounding rod verification.
- **Circuit Load Balancing:** High-load AC distribution testing.
- **Wire Gauge Audit:** Correct gauge for high-draw appliances.
- **Outdoor Fixture Rating:** Marine-grade/IP65 salt-air resistance.
- **Panel Access:** Logical location and bilingual labeling.
- **Surge Protection:** Whole-line 'Supresor de Picos' installation.
- **Elevator Brand & Service:** Local parts availability (Otis/Schindler vs. Generic).
- **Smart Home Integration:** CAT6 backbone wiring compatibility.

IV. Plumbing & Gas Safety

- **Pipe Material:** Pressure-tested PPR vs. standard PVC.
- **Hydropneumatic Pressure:** Flow testing on highest floors.
- **Water Softening (Suavizador):** Calcium protection systems.
- **Gas Line Ventilation:** Carbon monoxide risk assessment for heaters.
- **Pressure Relief Valves:** Boiler/heater explosion prevention.
- **Sewage Venting:** Methane stack check to prevent 'smell-backs'.
- **Shut-off Valve Mapping:** Individual room isolation capability.
- **Reverse Osmosis Ready:** Under-sink filtration space/plumbing.

V. Finishing & Architectural Detailing

- **Paint/Finish Quality:** Chukum application vs. limestone imitation.
- **Tile Lippage:** Floor leveling and trip hazard audit.
- **Joinery & Carpentry:** Tropical hardwood (Tzalam/Parota) vs. MDF.
- **Hurricane Glass:** Laminated glass thickness and impact rating.
- **Acoustic Insulation:** Decibel testing between condo walls.
- **Cabinet Hardware:** Stainless steel hinges vs. corrosive standard steel.
- **Sliding Door Tracks:** Recessed/flush mounting quality.
- **Pool Chemistry System:** Saltwater vs. chlorine ventilation.

VI. Legal & Regulatory Foundations

- **Manifestación de Obra:** Final 'Termination of Work' permit.
- **DRO Audit:** Verification of the Director Responsable de Obra.
- **Zoning Compliance:** Floor count vs. local SEDETUS regulations.
- **Environmental Impact (MIA):** Jungle/Beach-front specific permits.
- **Federal Zone Status:** Beachfront private property boundary audit.
- **Fire Safety:** Smoke detector and extinguisher common area audit.
- **Emergency Egress:** Unlocked, direct-to-street fire stairwells.
- **Cistern Capacity:** Water storage reserves for grid outages.
- **Internet Infrastructure:** Fiber optic last-mile connectivity.
- **The 'Sniff' Test:** Developer track record and past project finishes.

Integrity Clause: This audit is an architectural assessment of visible and documented systems. It does not replace a formal structural engineering survey but identifies the red flags that standard real estate agents routinely miss.